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Compulsory assignment

CDS Security & infrastructure

Case: Kvist & Bjælke ApS

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Introduction

Kvist & Bjælke ApS is a company that has wanted to develop a new login platform with a strong focus on security for some time.

The company wants to ensure that users' passwords cannot be read in plaintext, even if unauthorized individuals gain access to the system's database.

Kvist & Bjælke ApS has also decided to modernize its IT infrastructure. The company wants a more structured network architecture where different types of devices and systems are appropriately organized within the network.

The new login platform must therefore not only be developed but also deployed and made accessible through the company's server infrastructure.

You have been assigned the task of developing the login platform and designing and establishing the necessary infrastructure.

The assignment is divided into two courses:

1. **CDS.security:** Development and security assessment of the login platform.
2. **CDS.infrastructure:** Network design, IP planning, virtualization, and hosting of the login platform.

The two courses are interconnected, as the solution developed in CDS.security must subsequently be deployed on the infrastructure established in CDS.infrastructure.

Part 1 – CDS.security

1.1 Development of the login platform

Kvist & Bjælke ApS wants a login platform where security is considered from the beginning of the development process.

The platform must meet the following minimum requirements:

- Users must be able to authenticate using a username and password.
- The platform must implement role-based authorization with the roles `user`, `superuser`, and `admin`. Each role are specified below
- **User role:** Should only have permission to read users in a table.
- **SuperUser role:** Should have permission to create, read, update and delete users.
- **Admin role:** Should be able to create, read, update and delete users, superusers and admins.

- Users' passwords must never be stored or accessible in plaintext.
- Passwords must be stored using a secure password-hashing algorithm.
- The login platform must be accessible as a web-based application.
- The solution must be capable of running inside a Docker container.

You are free to choose the technologies and frameworks used to develop the platform.

You must be able to justify your choices, particularly regarding security.

1.2 Implementation

Based on the description you must develop the login platform.

You must ensure that the solution meets the specified requirements.

During development, you must document the most important security-related decisions.

The documentation may include:

- A description of the chosen authentication method.
- A description of how passwords are handled and stored.
- Screenshots demonstrating key functionality.

The source code must be stored in a GitHub repository.

Part 2 – CDS.infrastructure

Kvist & Bjælke ApS wants to establish a structured IT infrastructure that supports both the company's daily operations and the newly developed login platform.

The company requires a network architecture where employees, servers, and other devices can communicate efficiently and appropriately.

You must therefore propose a network design for the company and subsequently implement a smaller part of the infrastructure in a virtual environment.

2.1 The Company's network

Kvist & Bjælke ApS has several employees and IT systems that must operate within the company's network.

The company currently has the following devices:

Device	Quantity	Description
Desktop computers	15	Workstations used for administration and office tasks
Laptops	10	Used by employees working from different locations
Network printers	3	Shared printers available to employees
Servers	3	Hosting internal systems and applications
IP phones	12	Telephony using the company's network
Wireless access points	4	Wireless network access for employees and guests
Surveillance cameras	6	IP-based video surveillance
Network equipment	As required	Switches, routers, and other necessary network equipment
Guest devices	Variable	Smartphones and computers belonging to visitors

The devices listed above represent the company's current requirements.

You must also consider that the company may expand and introduce additional employees and devices in the future.

2.1.1 Development of an IP-addressing plan

You must develop an IP addressing plan for Kvist & Bjælke ApS.

The IP addressing plan must provide an overview of how IP addresses are distributed among the company's different devices and systems.

At a minimum, the IP addressing plan must include:

- Division of the company's network into appropriate logical networks.
- Selection of private IP address ranges.
- Specification of subnet masks or CIDR notation.
- Identification of the default gateway for each network.

- Description of which types of devices belong to each network.
- Considerations regarding static IP addresses and DHCP.

You must justify your choices regarding network segmentation.

For example, consider whether the company's servers, employee computers, printers, IP phones, and guest devices should be connected to the same network.

Remember that you are already given the VLAN 570 and 571. 570 is the client VLAN -> 10.5.70.0/24. 571 is the server VLAN -> 10.5.71/24

2.1.2 Subnetting

As part of your IP addressing plan, you must demonstrate a basic understanding of subnetting.

You must:

- Select at least two networks from your IP addressing plan.
- Specify the network address, subnet mask, and number of available host addresses.
- Explain why the selected subnet sizes are appropriate for the number of devices.
- Consider whether the subnets allow for future expansion.

Extensive subnet calculations are not required. The purpose is to demonstrate that you understand the relationship between subnet size and the number of devices on a network.

2.2 Establishing a server on VLAN 571

The login platform developed in CDS.security must now be deployed on the company's server infrastructure.

You must create a virtual machine with the following specifications:

Parameter	Requirement
Operating system	Ubuntu Server
Network	VLAN 571
Virtualization	Virtual machine
Application	Login platform developed in CDS.security
Hosting	Docker container
Network access	TCP port within the range 8000–8100

The virtual machine must be installed with Ubuntu Server.

After installing the operating system, you must install Docker and prepare the server to run the login platform.

The login platform must then be deployed and executed inside a Docker container.

You must ensure that the application is accessible using the server's IP address and the selected port.

You must select a port within the range **8000–8100** and document your choice.

Using Docker Compose is permitted.

2.3 Establishing a client on VLAN 570

Kvist & Bjælke ApS wants employees to access the login platform from a different network than the one hosting the server.

You must therefore create an additional virtual machine.

This virtual machine must be connected to **VLAN 570**.

The machine will function as a client and must be used to test connectivity to the login platform.

You must:

1. Create a virtual machine connected to VLAN 570.
2. Configure the machine's network settings.
3. Verify that the machine has the expected IP configuration.
4. Investigate whether the client can communicate with the login platform on VLAN 571.
5. Access the login platform using the selected TCP port between 8000 and 8100.
6. Document that the login platform functions correctly when accessed from the client machine.

Firewall and communication between VLANs

The network firewall has already been configured to allow the necessary communication from VLAN 570 to VLAN 571 on TCP ports 8000–8100.

You are therefore not required to configure any rules on the network firewall yourself.

However, you must be able to explain:

- Why VLAN 570 and VLAN 571 are separate networks.
- Why communication between VLANs typically requires routing.
- The role of the firewall in controlling communication between networks.
- Why allowing access only to specific ports can be beneficial.

2.5 Network Design (Optional)

Kvist & Bjælke ApS wants a visual overview of its future network infrastructure.

You must therefore create a network diagram illustrating an appropriate network architecture for the company.

At a minimum, the network diagram must include:

- Internet connectivity.
- Router and firewall.
- Relevant network switches.
- The company's different networks and VLANs.
- Server placement.
- Employee computers and other relevant devices.
- Wireless networks.
- Connections between the different network segments.

The diagram must clearly illustrate how you propose to structure the company's network.

You may use tools such as Draw.io or similar applications.

Note: The network diagram must illustrate your overall proposal for the company's network infrastructure. You are not required to implement all the networks and devices shown in the diagram.

2.6 Testing and Documentation

Once the infrastructure has been established, you must document that the solution works as intended.

The documentation must demonstrate the following:

Server – VLAN 571

- Ubuntu Server has been installed.
- The server has the correct IP configuration.
- Docker has been installed and is functioning.
- The login platform is running inside a Docker container.
- The login platform is accessible through the selected port.

Client – VLAN 570

- The client has the correct IP configuration.
- The client can access the login platform on VLAN 571.
- The login platform's functionality can be used from the client machine.

You must also briefly describe any problems encountered during the implementation and explain how you resolved them.

Part 3 – Final submission

This is a group assignment, and you must submit one combined "report" as the ReadMe file for your GitHUb repository covering both courses. It should show relevant screenshots and needed info.

The report must be clearly divided into sections for CDS.security and CDS.infrastructure. ## CDS.security – Submission requirements

Your submission must include:

1. Description of the developed login platform.
2. Documentation of the implemented security features, together with alternatives you find along the way.
3. Link to the GitHub repository containing the source code.

CDS.infrastructure – Submission requirements

Your submission must include:

1. IP addressing plan for Kvist & Bjælke ApS.
2. Brief documentation demonstrating subnetting.
3. Network diagram illustrating the company's proposed infrastructure.
4. Documentation of the Ubuntu Server installation and configuration on VLAN 571.
5. Documentation of the Docker installation and deployment of the login platform.
6. Documentation of the client machine on VLAN 570.
7. Documentation demonstrating communication between the client and the login platform.
8. Brief description of any challenges encountered and how they were resolved.

Final reflection

Conclude your submission with a brief overall reflection on the assignment.

Consider questions such as:

- How are security and infrastructure interconnected?
- What technical challenges did you encounter?
- What did you learn about deploying an application on a server?
- What would you do differently if the solution were to be deployed in a real company?

Formal requirements

- The assignment must be completed in groups (2 - 4 students) Rememeber to add them on Moodle.
- Both courses must be included in the submission.
- The “report” is as well-documented ReadMe available on the Github repository
- Documentation may be supported by screenshots, tables, and diagrams.
- The source code must be available through GitHub.
- The assignment must be submitted through Moodle.

Submission deadline: 6th of November 2026 @ 23:59 on Moodle